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ملاحق

كلية الهندسة المعلوماتية

السنة الثانية

ملحق أول محاضرتين Signals

عمل طلابي



RB Informatics; 12/06/2026

الاتصالات الرقمية

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In this handbook, we will have more explanation of topics in the first two lectures beside Q&As for each topic.

You can read the explanation, solve the questions or both simultaneously

Lecture "1"

Core words in the lecture:

1. Digital: Refers to electronic technology (devices that work with electricity (like computers, smartphones, and routers)) that generates data (creating it), stores it (save it on hard drives or flash drives) and processes (calculates or changes it) using binary code—a series of 0s and 1s (bits).

Example: When you type the letter "A" on a keyboard, the computer processor does not see a geometric letter. Instead, your electronic technology immediately translates that stroke into an exact, predefined sequence of bits: 01000001. The machine stores and processes that exact string of binary code to draw it on your screen.

2. Communication: Is the process of exchanging information, ideas, thoughts and meaning (sharing a message cleanly) between individuals or groups (senders and receivers). In network communication, the communicating entities are applications (software programs running on the devices).





Example: When a WhatsApp message is sent, the communication is not exactly between two people or two mobile phones; it is between the WhatsApp application on the sender side and the WhatsApp application on the receiver side. The physical plastic phone is just a container, while the application software formats, handles, and receives the message rules.

3. Digital communication: Is the transfer of information, data, or messages using digital signals (pulses representing bits), typically encoded in binary (0s and 1s) and transmitted over electronic networks, including internet, wireless or storage channels (like fiber cables, Wi-Fi waves, or hard drives).

Example: When you record an audio note on your phone and hit send, your phone does not send your actual continuous voice through the air. Instead, it converts your voice into numbers, encodes those numbers into binary 0s and 1s, and then transfers those bits as a rapid series of electrical or radio pulses over the internet network straight to your friend.

4. Communication system: Is a collection of components including a transmitter (or a source that converts originated data into a signal suitable for transmission), channel (the media/medium through which data is sending like copper wires, fiber optics, or radio waves) and receiver (converts the received signal back into data) that work together to exchange information, such as voice, data, or video, from a source to a destination.

Example: Think of watching a live video stream on YouTube. The camera acts as the **Source** capturing raw data; the laptop's network hardware acts as the **Transmitter** changing pixels into signals; the invisible air waves or physical internet lines act as the **Channel** carrying it; your phone's internal antenna acts as the **Receiver** catching and unpacking the pulses; and finally, your YouTube app screen acts as the **Destination** showing you the smooth video.

5. Signal: Is something that carries information. It is a physical quantity (a real-world, measurable entity like voltage on a wire, electric current, or electromagnetic waves) that can be stored (saved on hardware), manipulated (changed or reshaped by physical processes) and transmitted.

Example: If you are stranded on an island and use a mirror to reflect sunlight toward a rescue ship using Morse Code, the letters of your message are the data. The **flashes of light** hitting the ship are the **Signal**. It is a physical quantity (light waves) that you manipulated (blocked and released) and transmitted across space to carry your text.



- Q&As -

1. The receiver must convert:

- a) The receiver data into signal
- b) The received signal into data
- c) The received signal into digital data
- d) The received analog data into analog signal

Correct: b

- Because the transmitter sends a signal the receiver's job is to change that signal back into original data.

2. Transmitter Converts originated data into:

- a) Suitable form for transmitting (signal)
- b) Digital form
- c) Analog form
- d) B+C

Correct: a

- Because transmission media need a signal, not just digital or analog form-
The transmitter converts data into a signal suitable for them.

**REMEMBER
WHO
YOU
WANTED
TO BE**



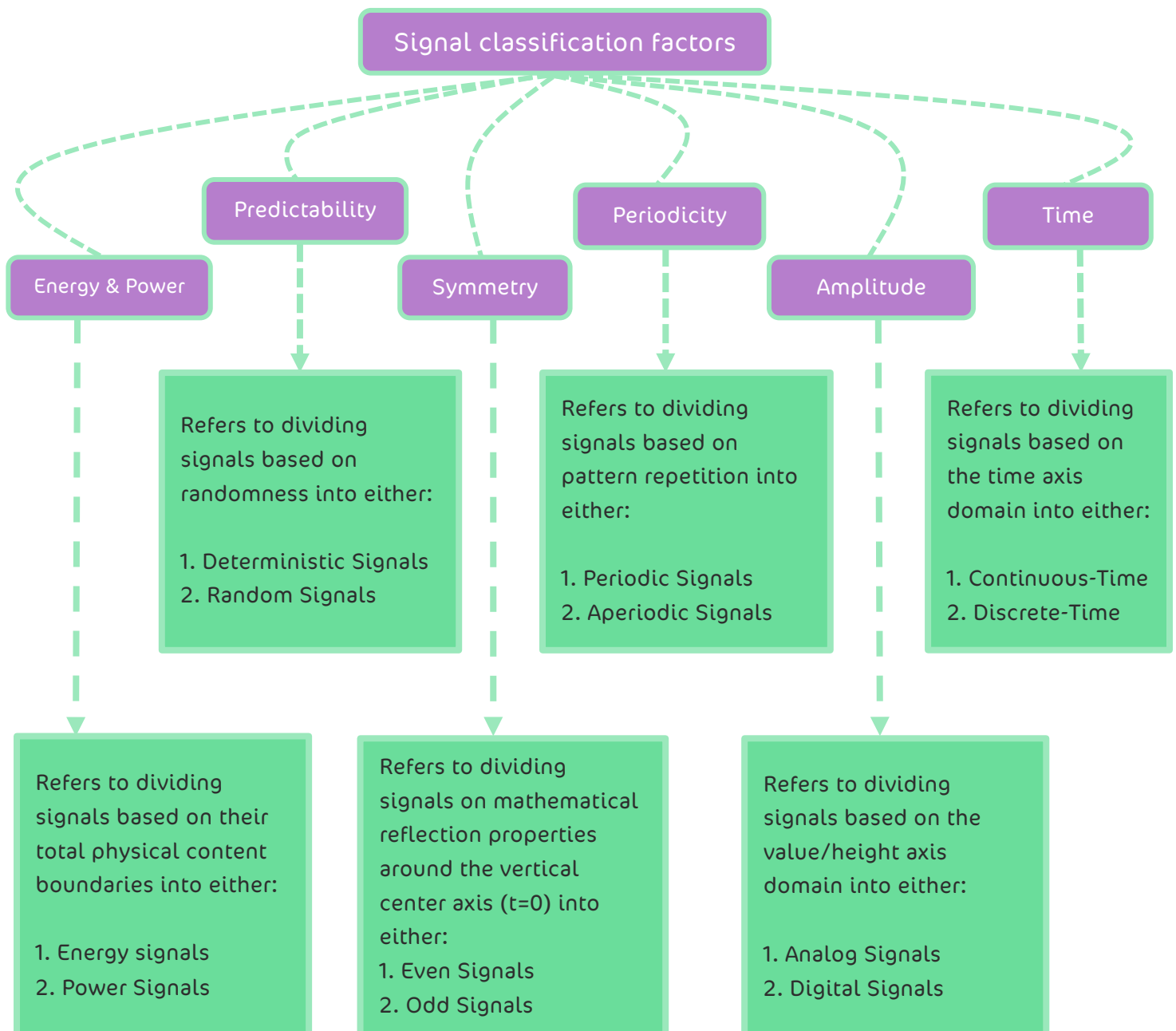
Signal classification factors

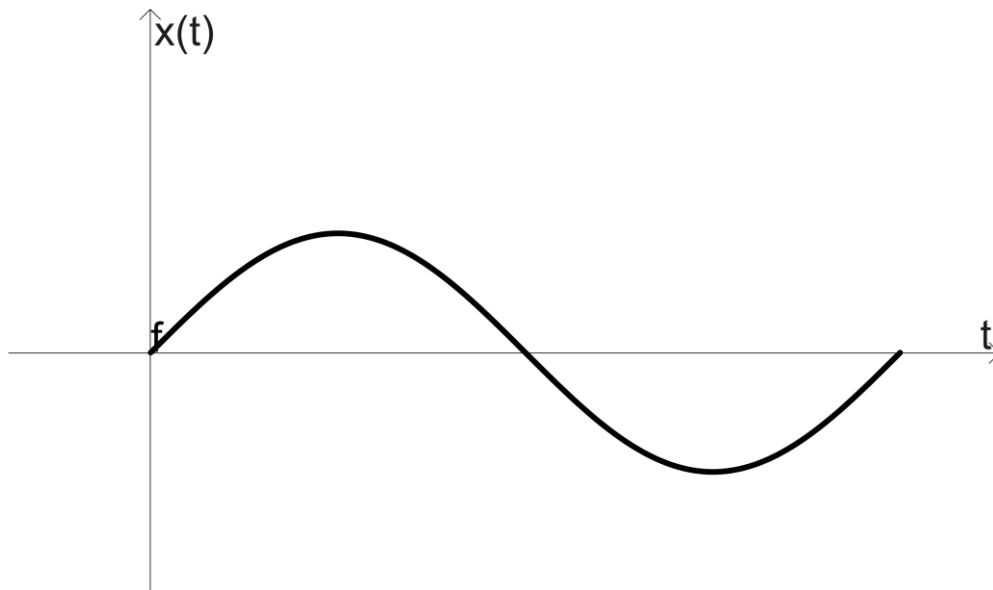
We classify signals because different applications and systems deal with different types of signals. Each application may work with a specific type of signal.

BRANCH 1: Time

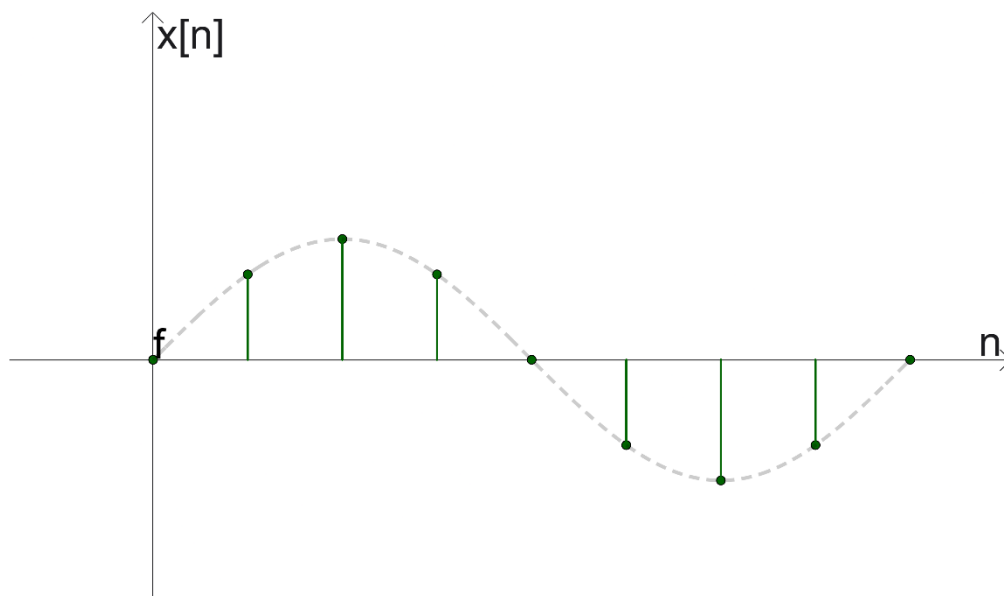
Refers to dividing signals based on the time axis domain into either:

Continuous-Time: represented as $x(t)$, meaning the signal is unbroken and has a valid value at every single microsecond of time.





Discrete-Time: represented as $x[n]$, meaning the signal is a series of independent samples defined only at specific, isolated integer steps.



Example: Think of your human voice traveling through the air; it is continuous because it flows smoothly without any pauses or empty gaps. However, when a computer records that voice into a digital music file, it takes snapshot "samples" thousands of times per second—creating a discrete sequence where the computer knows nothing about the space between those samples.

**Don't wish for it,
Work for it 🔥**



Sampling: how to obtain $x[n]$ from $x(t)$:

Sampling is the process of converting a continuous-time signal $x(t)$ (defined at every unbroken microsecond of time t) into a discrete-time signal $x[n]$ (defined only at isolated integer sample points n) by substituting the continuous time variable (t) with $t = n * T_s$, where (n) is an integer index ($\dots, -2, -1, 0, 1, 2, \dots$) and (T_s) is the **Sampling Period** (the fixed, uniform time interval between consecutive snapshots).

Example: Imagine you given by $x(t) = 3 \cos(2\pi t)$ for a signal, and a computer system takes a snapshot sample of it exactly every $T_s = 0.25$ seconds. **[Knowing that for any given signal formula: $x(t) = A * \cos(2\pi t)$ && $f = 1$]**

To find out what the computer actually sees, you remove the (t) and plug in ($n * 0.25$):

At sample $n = 0$: $t = 0 * 0.25 = 0$ seconds $\rightarrow x[0] = 3 \cos(0) = 3$

At sample $n = 1$: $t = 1 * 0.25 = 0.25$ seconds $\rightarrow x[1] = 3 \cos(2 \pi 0.25) = 3 \cos(0.5\pi) = 0$

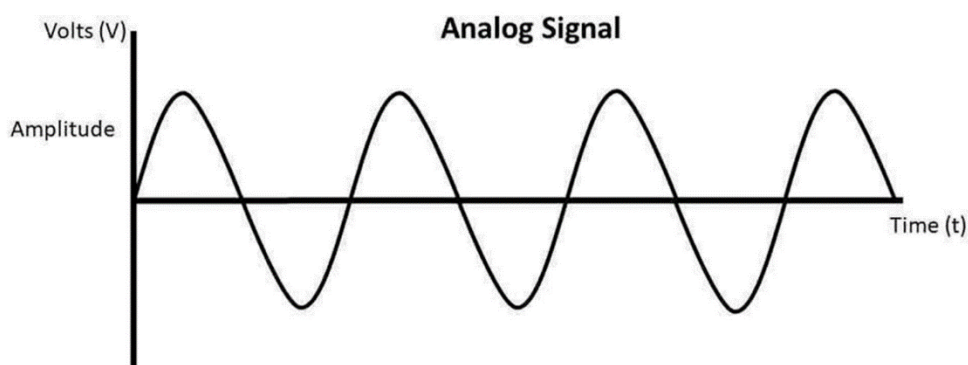
At sample $n = 2$: $t = 2 * 0.25 = 0.5$ seconds $\rightarrow x[2] = 3 \cos(2\pi 0.5) = 3 \cos(\pi) = -3$

The electronic system discards the continuous wave and saves only this isolated sequence of numbers: $\{3, 0, -3, \dots\}$, creating the discrete-time signal $x[n]$.

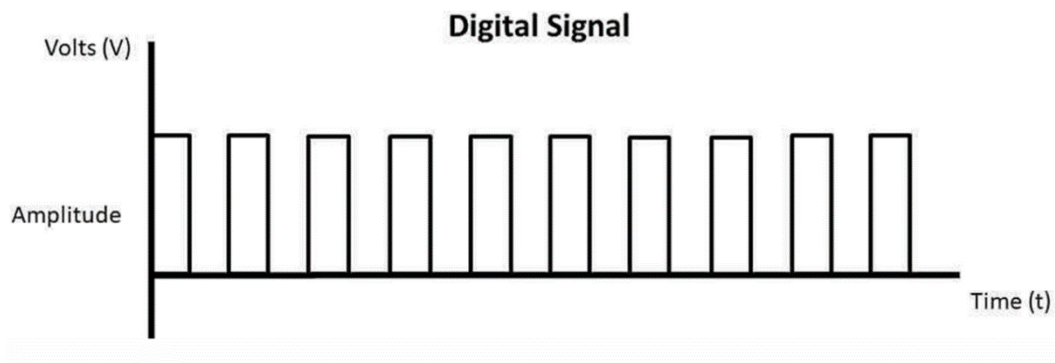
BRANCH 2: Amplitude

Refers to dividing signals based on the value/height axis domain into either:

Analog Signals: where the signal's strength/intensity (volts [v]) can take any infinite real value within a continuous range.



Digital Signals: where the intensity maintains a strict constant level for a period and then changes to another constant predefined value, representing discrete levels like binary.



Example: Think about adjusting the volume of a speaker. An analog volume knob lets you turn it smoothly with your hand, changing the sound continuously to absolutely any level (like 15.3% or 15.34%).

However, a digital volume button setup works in strict steps; when you press it, the volume level jumps directly from level 15 to level 16, without letting you stop anywhere in between.

- Q&A -

1. Signal $x(t)$ can be continuous-time, (t) is independent variable:

- a) $x(t)$ can take on any value in the continuous interval (a, b) , then $x(t)$ is discrete.
- b) $x(t)$ can take on any value in the continuous interval (a, b) , then $x(t)$ is analog.
- c) $x(t)$ can take on some values in the continuous interval (a, b) , then $x(t)$ is analog.
- d) $x(t)$ can take on any value in the continuous interval (a, b) , then $x(t)$ is digital.

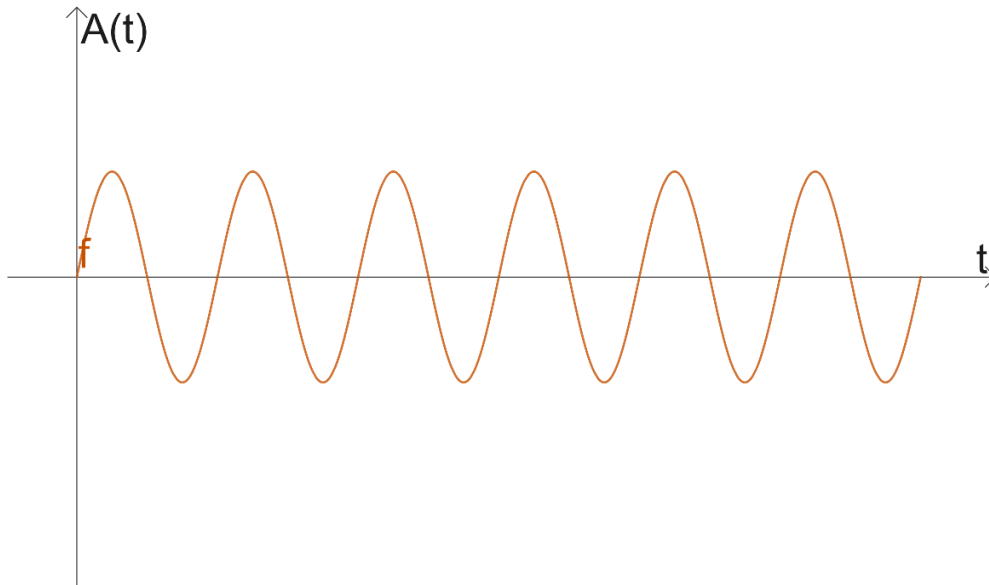
Correct: b

Because if the signal can take any value within a continuous range, this is the definition of an analog signal.

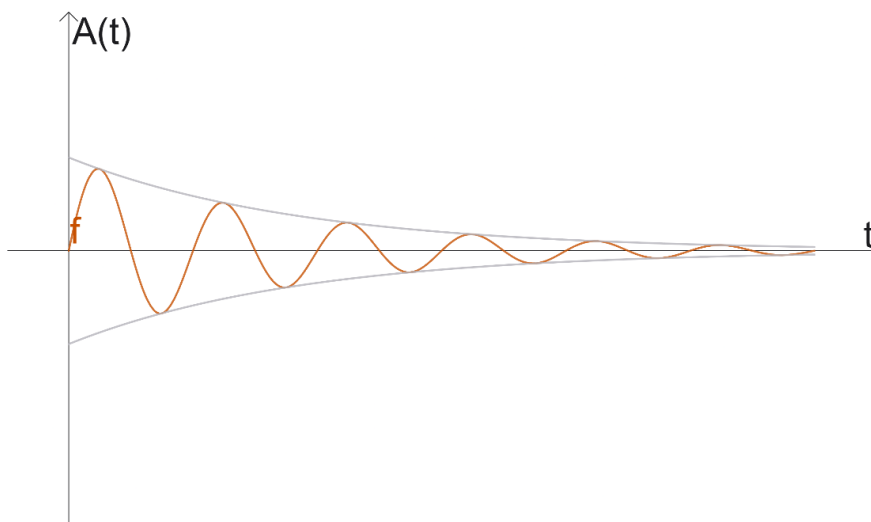
BRANCH 3: Periodicity

Refers to dividing signals based on pattern repetition into either:

Periodic Signals: which repeat a perfect, identical waveform pattern over a constant fixed duration of time called the Period, satisfying $x(t + T) = x(t)$.



Aperiodic Signals: non-periodic waveforms that have changing, unique patterns over time and never repeat identically.



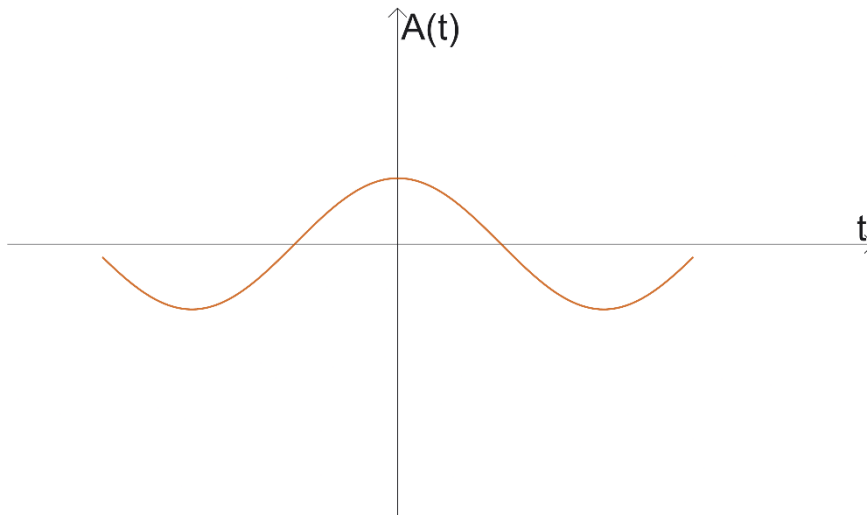
Example: An electronic wall clock oscillator or a perfect musical tuning fork wave is a periodic signal because it swings back and forth over the exact same time interval over and over. Your conversational speaking voice, on the other hand, is completely aperiodic because you constantly change your words, pacing, and sounds without any recurring pattern.



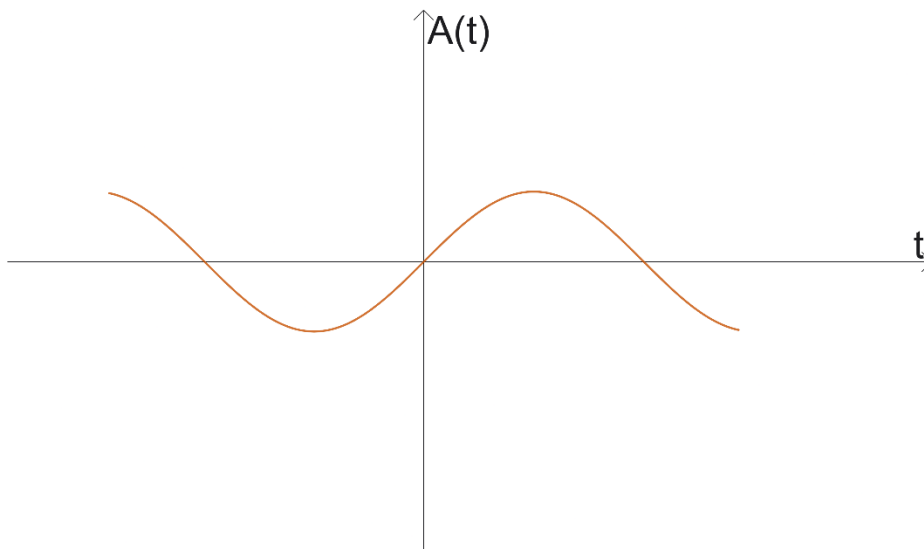
BRANCH 4: Symmetry

Refers to dividing signals based on mathematical reflection properties around the vertical axis or the center into either:

Even Signals: symmetrical mirror images satisfying $x(-t) = x(t)$ or $x[-n] = x[n]$



Odd Signals: antisymmetric inverted shapes satisfying $x(-t) = -x(t)$ or $x[-n] = -x[n]$.



Example: Think of a perfect Cosine Wave $x(t) = \cos(t)$ is an Even Signal, whereas the perfect Sine Wave $x(t) = \sin(t)$ is an Odd Signal.

فكك مني رايم اشرب قهوة ☕





BRANCH 5: Predictability

Refers to dividing signals based on randomness into either:

Deterministic Signals: signals whose future values can be computed exactly ahead of time using a known mathematical formula with 100% certainty.

Random Signals: unpredictable waveforms whose future values cannot be determined by an equation and must be analyzed using probability.

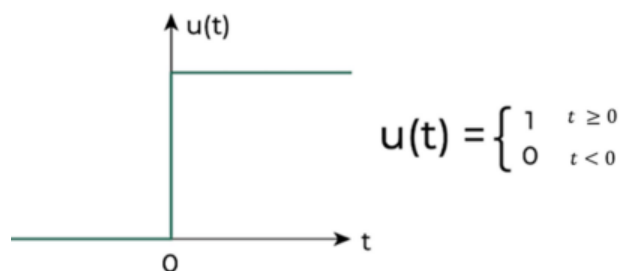
Example: A standard AC electrical power outlet signal $[220V \cdot \sin(2\pi \cdot 50 \cdot t)]$ is deterministic because an engineer can calculate its exact position 10 hours from now using math. The static **background hissing noise** coming out of an un-tuned radio is a random signal because it is a chaotic mess of atomic interference that is completely unpredictable.

BRANCH 6: Energy & Power

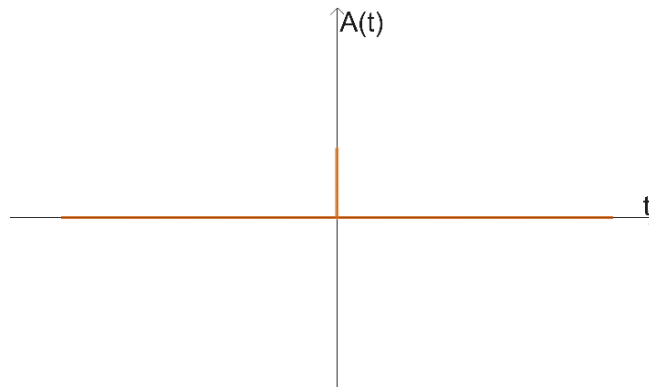
Refers to dividing signals based on their total physical content boundaries into either:

Energy Signals: signals that have a finite total energy value ($0 < E < \infty$) but a long-term average power of zero ($p = 0$), usually because they are short pulse bursts.

Unit – step function Heaviside unit function



Power Signals: signals that have infinite total energy ($E = \infty$) but a stable, non-zero finite average power value ($0 < P < \infty$), typically because they are periodic or last forever.



-Q&As-

1. Data is an entity that convey some meaning based on some mutually agreed rules between the sender and the receiver and it can be:

- a) Analog or Digital.
- b) Continuous-time or Discrete-time.
- c) Analog continuous-time.
- d) Digital discrete-time.

Correct: a

Because Data can be analog or digital Continuous-time and discrete-time are classifications of signals, not data.

2. Signal $x(t)$ can be continuous-time, (t) is independent variable:

- a) $x(t)$ can take on any value in the continuous interval (a, b) , then $x(t)$ is discrete.
- b) $x(t)$ can take on any value in the continuous interval (a, b) , then $x(t)$ is analog.
- c) $x(t)$ can take on some values in the continuous interval (a, b) , then $x(t)$ is analog.
- d) $x(t)$ can take on any value in the continuous interval (a, b) , then $x(t)$ is digital.

Correct: b

Because if the signal can take any value within a continuous range, this is the definition of an analog signal [the definition of an analog signal].



3. $x(t)$ and $x[n]$ are even signals if:

- a) $x(-t) = x(t)$ and $x[-n] = x[n]$.
- b) $x(-t) = -x(t)$ and $x[-n] = -x[n]$.
- c) $x(-t) = x(-2t)$ and $x[-n] = x[-2n]$.
- d) A+B.

Correct: a

4. $x(t)$ and $x[n]$ are odd signals if:

- a) $x(-t) = x(t)$ and $x[-n] = x[n]$.
- b) $x(-t) = -x(t)$ and $x[-n] = -x[n]$.
- c) $x(-t) = x(-2t)$ and $x[-n] = x[-2n]$.
- d) A+B.

Correct: b

5. Signal $x[n]$ is defined at discrete times, n is an integer:

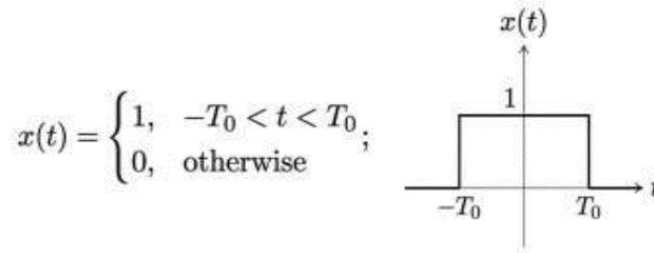
- a) Is discrete.
- b) $x[n]$ can take on any value in the continuous interval (a, b) , then $x(t)$ is analog.
- c) $x[n]$ can take on some values in the continuous interval (a, b) , then $x(t)$ is analog.
- d) $x(t)$ can take on any value in the continuous interval (a, b) , then $x(t)$ is digital.

Correct: a

Because n is an integer, so the signal exists only at separate time points $\rightarrow$ this is a discrete signal.



6. The signal is:



- a) Power signal - even - nonperiodic - deterministic
- b) Energy signal - odd - nonperiodic - random
- c) Power signal - even - periodic - random
- d) Energy signal - even - nonperiodic - deterministic
- e) Energy signal - odd - periodic - deterministic

Correct: d

Sinusoidal signal

$$x(t) = A \cdot \cos(\omega_0 t + \varphi)$$

A is the amplitude, which represents the maximum value of the sinusoidal signal.

W: radian frequency.

φ : Phase angle (radian)

$$T_0 = \frac{2\pi}{\omega_0} \quad , \quad f_0 = \frac{1}{T_0} \quad , \quad \omega_0 = 2\pi f_0$$

Benefit: A sinusoidal signal can be represented using either cosine or sine:

$$x(t) = A \cos(\omega_0 t + \varphi) \text{ or } x(t) = A \sin(\omega_0 t + \varphi).$$

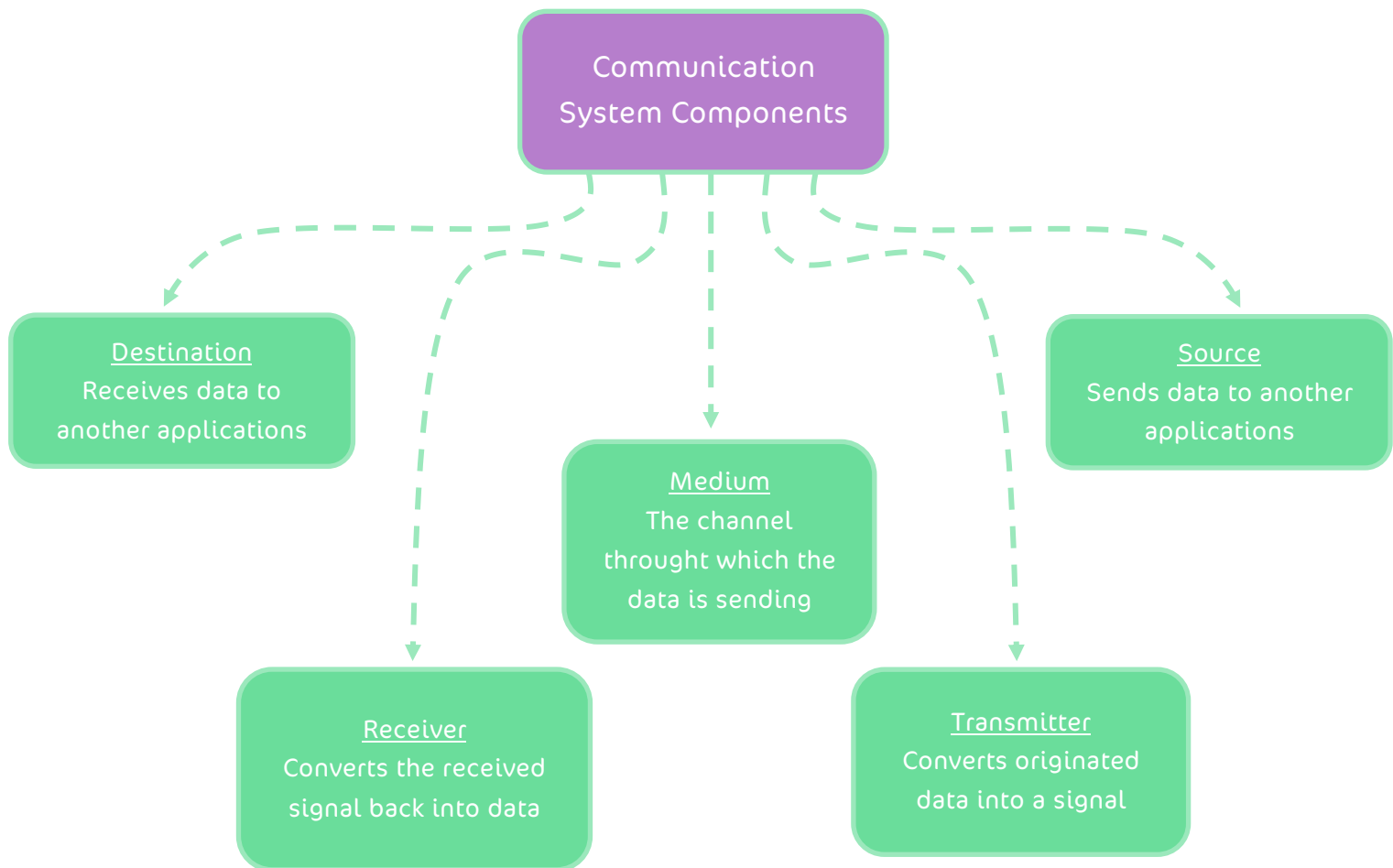
**YOUR FUTURE IS CREATED BY WHAT
YOU DO **TODAY**, NOT TOMORROW.**



Lecture "2"

Communications over net among applications:

As we said in the previous section, Signals are something that carries information from applications. These applications that send data are called a **Source**, they send data to another application which we call **Destination**.

**Note:**

Continuous signals are defined for a continuum of time (It's mathematical representation is $x(t)$), while discrete signals are defined only at specific discrete points in time (It's mathematical representation is $x[n]$).

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-Q&A-

Fill in the gaps;

Communication is a process of data from on point which called somewhere and at some time to another point which somewhere during time

- a) Manipulation, Source, Computer, Minimum.
- b) Collecting, Source, Computer, Acceptable.
- c) Transferring, Source, Destination, Acceptable.
- d) Collecting, Source, Destination, Defined.

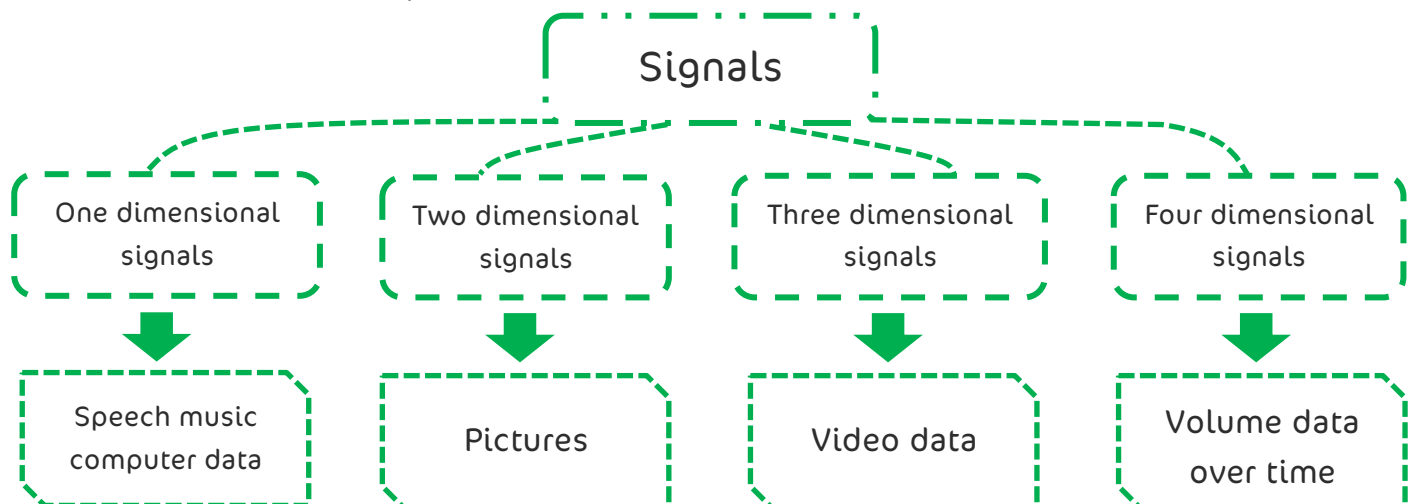
Correct: c

Signals and Dimensions:

Dimensions: refers to the number of independent variables (like time, horizontal space, vertical space, or frame movement) needed to define or calculate a signal's value.

In engineering math, a dimension is an axis or parameter that must change for us to see a change in the signal data.

- 1- **Dimension (1D):** The signal depends on only one variable (usually time).
- 2- **Dimensions (2D):** The signal depends on **two** variables (usually horizontal space 'x' and vertical space 'y').
- 3- **Dimensions (3D):** The signal depends on three variables (length, width, and movement over time).



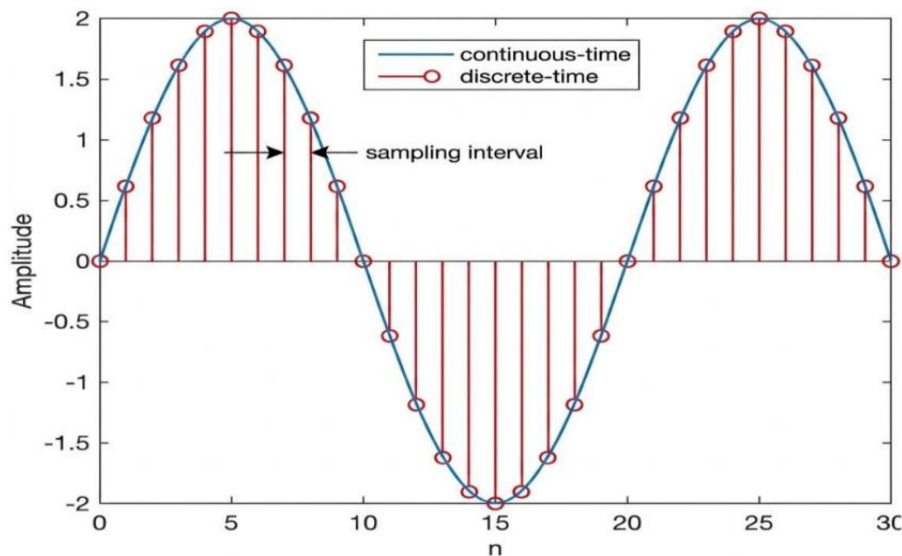
RBCs Note: Why do we need to know Dimensions of the signal?

We need to classify signals by dimension because different communication systems, processors, and software applications deal with different data structures. A system built to process a basic 1D audio wave cannot handle a complex 2D medical x-ray image or a 3D live movie stream. Knowing the dimension helps engineers select the right hardware chips, memory bandwidth, and mathematical processing formulas for that specific application.

Revisiting Signal Classifications:

1. Continuous-time VS Discrete-time:

Continuous-Time Signal: $x(t)$	Discrete-Time Signal: $x[n]$
Continuous signals are defined for a continuum of time (meaning the signal has a valid, measurable value at every infinite, unbroken instant of time t).	Discrete signals are defined only at specific discrete points in time (where n is a strict integer representing an index or sample number).



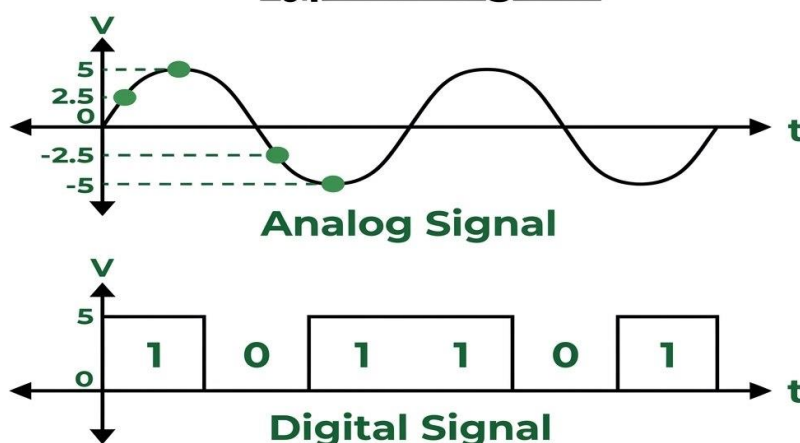
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2. Analog VS Digital:

Analog Signal	Digital Signal
Analog signals are continuous in time and amplitude (special case of continuous-time signals)	Digital signals are discrete in both time and amplitude (special case of discrete-time signals)
Wave-like, and represent data through varying amplitude	Are discrete, binary, and step-based (0's and 1's)
Analog is ideal for natural, real-world measurements Like sound and temperature	Digital excels in precision, storage, and noise resistance making it standard for modern computing, Data storage ..etc

Types of Signals



3. Periodic VS Aperiodic:

Periodic	Aperiodic
repeat their pattern at regular, fixed time intervals	Does not repeat itself and changes over time



NOTES:

Where do we see these kind of signals?

1. Analog Periodic signals:

- Radio broadcasting (AM/FM).
- Sensor Systems.

2. Analog Aperiodic signals:

- In communication systems for high rate transmission, natural and geophysical monitoring.

3. Discrete Periodic signals:

- DSP (Digital Signal Processing).
- Data Conversion.
- Communication Systems for sampling.
- Signal generation (sin wave).

4. Discrete Aperiodic Signals:

- DSP (digital filters).
- Control Systems (stability).
- Speech and image processing.



NOTE THAT:

FM: Frequency Modulation

AM: Amplitude Modulation

Modulation: It is the process in which the original signal is taken and placed on a carrier wave in order to transmit it (the transmitter does the modulation)

Demodulation: is the exact opposite operation of modulation, it takes the signal off the carrier wave.

DSP: like eliminating unwanted frequency components (such as noise) from a discrete signal, a digital filter is applied to attenuate those specific frequencies while allowing others to pass.



4. Energy vs Power signals:

<u>Power signals</u>	<u>Energy signals</u>
1- Normalized average power is finite and non zero ($0 < p < \infty$).	Normalize average energy is finite and non zero ($0 < E < \infty$)
2- Power of CT signal : $p = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)^2 dt$	Energy of CT signal : $E = \int_{-\infty}^{+\infty} x(t) ^2 dt$
3- Power of DT signal : $p = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n) ^2$	Energy of DT signal : $E = \sum_{n=-\infty}^{\infty} x(n) ^2$
4- Practical periodic signals are power signals.	Non periodic signals are energy signals.

Power signals:

- Periodic Signals: Any repeating waveform (sine, cosine, square, triangular) that continues indefinitely.
- Constant Signals: A non-zero signal that lasts forever, such as a DC signal.
- Step Function: A signal that changes value at a specific time and stays there ((t)).

Energy signals:

- Pulses: A single pulse or impulse that occurs once and dies out .
- Non-periodic Signals: Signals that do not repeat and exist only over a limited time intervals, such as a single decaying exponential ($e^{-at}u(t)$).
- Data and Images: Images, audio and similar data packets usually contain a finite amount of energy.





- Q&A -

1. The power signal is power if:

- a) The power is zero and the energy infinity.
- b) The power is finite and the energy is infinity.
- c) The power is limit zero for all the time.
- d) The energy is infinity all the time.

Correct: b

Power signal → Finite power + Infinite energy

Fundamental Signals:

Fundamental signals are basic, elemental waveforms—such as sinusoids, steps, impulses, and ramps — used to model and analyze physical complex systems and data.

These signals are considered the main tools in the world of Signals and Digital Communication.

These signals, often categorized by time continuity (continuous vs. discrete) or predictability (deterministic vs. random), are essential for understanding, representing, and manipulating data across time and frequency domains.

And they are:

- 1- Unit step functions $u(t)$ Heaviside function.
- 2- Unit Impulse function (Dirac Delta).
- 3- Unit Ramp function.
- 4- Exponential Signal.
- 5- Sinusoidal Signals.
- 6- Rectangular Pulse function.
- 7- Sinc function.

NOTE:

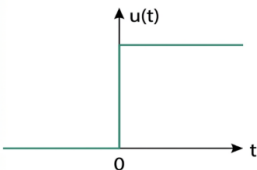
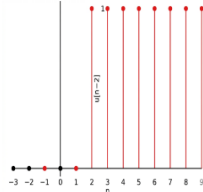
Function are the continuous Fundamental signals while sequence are the discrete Fundamental signals.

NOTE:

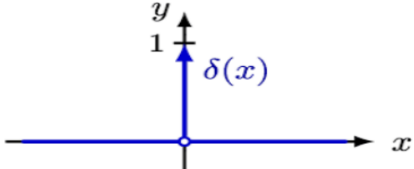
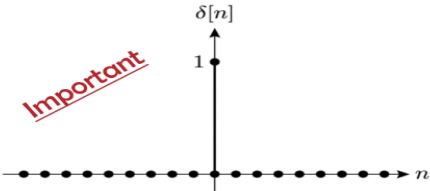
Know that we can do sampling to all continuous signals in order to have discrete sequence. Fundamental Discrete signals are essential for digital signal processing, control, and communication systems

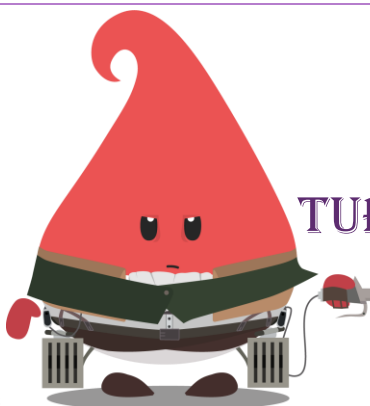
We will make tables to manage all kinds of these signals (continuous vs. discrete):

Unit Step Function U(t) vs Unit Step Sequence:

Unit (Heaviside) Step Function	Unit Step Sequence
is a piecewise function that equals (0) for negative arguments and (1) for positive arguments, representing a switch that activates at $t = 0$. Defined as $u(t) = 0$ for $t < 0$ and $u(t) = 1$ for $t \geq 0$,	is a fundamental discrete-time signal that acts as a step function, having a value of 0 for all negative indices ($n < 0$) and a value of 1 for all non-negative indices ($n \geq 0$). It is defined as $u[n] = \{..., 0, 0, 1, 1, 1, 1 ...\}$ where the arrow indicates the sample at $n = 0$. It represents the sudden application of a signal.
it is widely used in engineering to model systems and calculate Laplace transforms Usage: Switching on, Rectangular pulse	Primarily used in signal processing and system analysis as a standard input test signal, particularly to determine a system's stability and response to sudden changes.
 $u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$	 $u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$

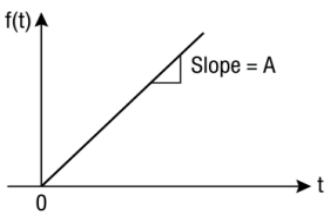
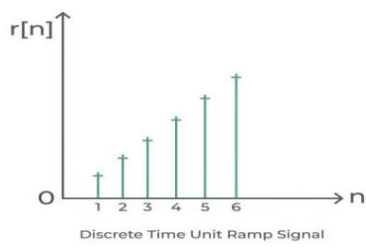
Unit Impulse Function vs Unit Impulse Sequence:

Unit Impulse Function <Dirac Delta>	Unit Impulse (sample) Sequence
	Is a fundamental discrete-time signal that equals 1 at $n = 0$ and 0 for all other values of n. It acts as an impulse at the origin and is essential for system analysis, convolution, and defining system impulse responses.
Usage: physics, signal processing.	System Testing: It is used as a standard input to observe transient responses and analyze the frequency response of filters. Signal Sampling: The unit impulse function is used to model the ideal sampling process, creating a discrete sequence from a continuous signal. Signal Processing & Edge Detection: In image processing, it can be used to model spatial characteristics
$\delta(x) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$	$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$
	



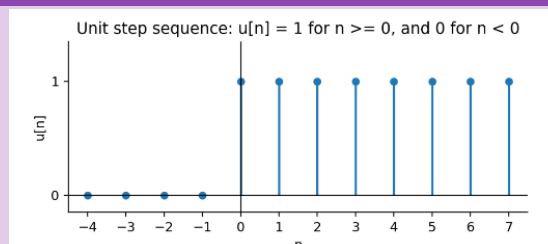
TURN YOUR HARD WORK INTO HISTORY!

Ramp Function $R(t)$ vs Unit Ramp Sequence:

Ramp Function	Unit Ramp Sequence
	Is a discrete-time signal that starts at zero and increase linearly with a slope of one for non-negative indices. It is defined as $r[n] = n \cdot u[n]$ which means $r[n] = n$ for $n \geq 0$, and $r[n] = 0$ for $n < 0$. It acts as a foundational signal in digital signal processing and control systems.
Usage: control systems, signal processing, power supply.	-Testing systems: Used to check the response of digital systems to linearly increasing inputs. -Modelling: Represents inputs that grow linearly over time.
$R(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$	$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$
	 Discrete Time Unit Ramp Signal

For the below shape, choose the correct signal name:

- A. Unit step sequence
- B. Unit impulse sequence
- C. Unit impulse function
- D. Unit step function



Correct: A

Operations on Signals (on time):

The operations on Signals are usually used to manipulate time and amplitude
But for now we will learn operations that affects time.

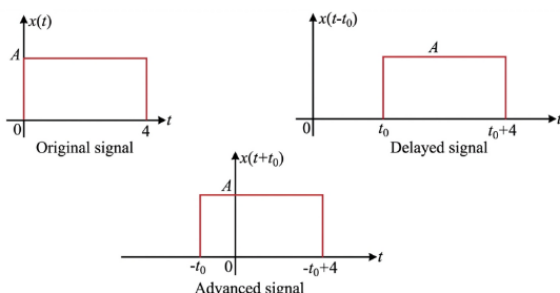
Time shifting

Moves the signal on the time axis back (delay) and forth (advance)

Does not affect neither amplitude nor shape

represented as $y(t) = x(t \pm t_0)$

- Negative Shift ($t - t_0$): Delays the signal, moving it right (time delay).
- Positive Shift ($t + t_0$): Advances it left.



This operation is a core building block for various engineering and analytical applications.

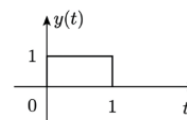
Synchronization and Alignment: It is used to align multiple audio or data tracks so they occur simultaneously, which is critical in multimedia production and multi-sensor data fusion.

Time scaling

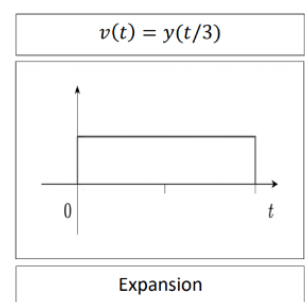
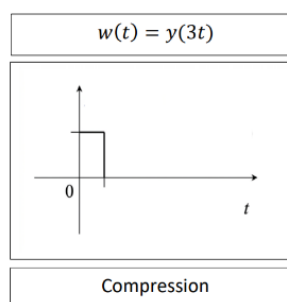
compresses or expands a signal $x(t)$ by multiplying its time variable by a non-zero constant α , resulting in $y(t) = x(\alpha t)$

- If $|\alpha| > 1$, the signal is compressed (speeds up)
- If $|\alpha| < 1$, it is expanded (slows down)

the amplitude remains unchanged.



The solution:



Time shifting and scaling

1. Scale First: Perform the time scaling ($t \rightarrow \alpha t$) first.
2. Shift Second: Perform the time shift ($\alpha t \rightarrow \alpha t - b$) second.

Note for time shifting:**How Systems Handle Advanced Signals (Time Advance)??**

Definition: refers to the practical engineering method used when a math formula shifts a signal forward in time (a time advance). Since a physical system cannot see the future, it temporarily uses a stored or predicted value to keep working until the real signal arrives, then updates itself based on the difference.

Deep Explanation: Mathematically, we can shift a signal anywhere on paper. But physically, a system cannot look into the future or use a piece of data before it arrives. To solve this incorrect physical reality, engineers use a clever solution:

The Guess (Prediction): The system looks at previous saved values of the signal and takes a guess (treats a past or estimated value as if it were the true upcoming value).

The Processing: The system continues running its operations using this temporary placeholder value.

The Correction (Adjustment): The microsecond the true data value actually arrives, the receiver application performs an instant mathematical comparison between the real value and the guessed value. If there is a mistake, the system immediately adjusts its rules to match the true incoming data perfectly.

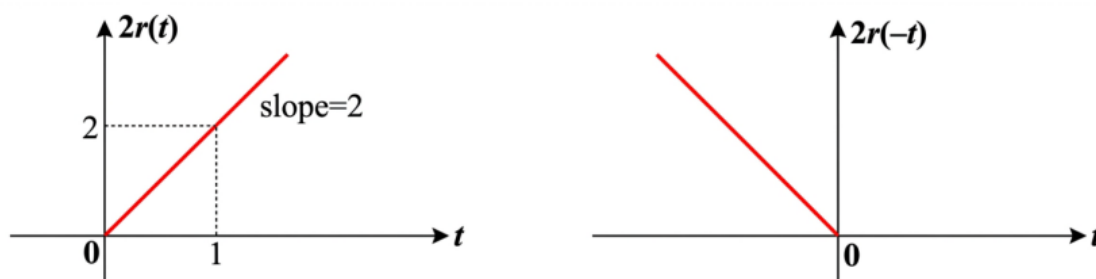
Time Reversal (with a ramp signal):

Refers to flipping a signal horizontally across the vertical center axis (the amplitude axis) to create a mirror image, which mathematically replaces the time variable t with negative time $-t$. For a standard ramp signal $r(t) = t u(t)$, this transformation results in $r(-t) = -t u(-t)$.

Deep Explanation: Think of time reversal like hitting the "rewind" button or looking at a signal through a mirror placed on the vertical y-axis.

The Original Signal ($r(t)$): A ramp signal is like a straight ramp going up. It starts at time zero ($t=0$) and climbs up linearly as time moves forward into the positive future. For negative time, it is completely dead (zero).

The Flipped Signal ($r(-t)$): When we apply time reversal, everything swaps sides horizontally. What used to happen in the future now happens in the past! The ramp is mirrored over to the left side. This means the signal now lives only in negative time. As time moves from the past toward zero, the signal slants down and ends perfectly at $t=0$.



- Q&As -

1. $Y(t) = X(2t - 3)$ means:

- A. Advance in 2 and compress with 3
- B. Delay in 3 and compress with 2
- C. Time scaling in 3 and stretch with 2
- D. Advance in 3 and stretch with 2
- E. Delay in 2 and stretch with 3

Correct answer: B - Delay in 3 and compress with 2.

2. $y(t) = x(at)$ means:

- A. Amplitude scaling
- B. Time scaling
- C. Multiplication of signals
- D. None

Correct answer: B - Time scaling.

Inter-relationships:

Note: Relationships to memorize + read their conclusion

Continuous Time:

- $u(t) = \int_{-\infty}^t \delta(\tau) d\tau$
- $\delta(t) = \frac{du(t)}{dt}$

Discrete Time:

- $u[n] = \sum_{k=0}^{\infty} \delta[n-k]$
- $\delta[n] = u[n] - u[n-1]$

U(t) & R(t):

- $r(t) = \int_{-\infty}^t u(t) dt = \int_{-\infty}^0 u(t) dt + \int_0^t u(t) dt = 0 + \int_0^t u(t) dt = t;$

$$\begin{aligned} r(t) &= 0 \text{ for } t < 0 \\ &= t \text{ for } t > 0 \end{aligned}$$

U[n] & R[n]:

- Unit Step Sequence (u[n]):

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

- Unit Ramp Sequence (r[n]):

$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$r[n] = n \cdot u[n]$$

$$r[n] = \sum_{k=0}^n u[k]$$

$$u[n] = r[n] - r[n-1]$$

- Unit Impulse Sequence:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

- Unit Step Sequence:

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



- Unit Ramp Sequence:

$$r[n] = n \cdot u[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

From Impulse to Ramp (Accumulation):

- First summation (Impulse to Step): The running sum of an impulse is a step function.

$$\sum_{k=-\infty}^n \delta[k] = u[n]$$

- Second summation (Step to Ramp): The running sum of a step function is a ramp function.

$$r[n] = \sum_{j=-\infty}^n u[j] = \sum_{j=0}^n \left(\sum_{k=0}^j \delta[k] \right)$$

- Therefore:

$$r[n] = \sum_{k=0}^n u[k]$$

From Ramp to Impulse (Difference):

- First Difference: $r[n] - r[n-1] = u[n]$ (the slope changes only at $n=0$).
- Second Difference: $u[n] - u[n-1] = \delta[n]$.
- Substituting $u[n]$ gives the relationship:

$$\delta[n] = r[n] - 2r[n-1] + r[n-2]$$





<u>Signal Relationship</u>	<u>Formula</u>
Impulse to step	$u[n] = \sum_{k=0}^n \delta[k]$
Step to Ramp	$r[n] = \sum_{k=0}^n u[k]$
Impulse to Ramp	$r[n] = \sum_{k=0}^n \sum_{m=0}^k \delta[m]$
Ramp to Impulse	$\delta[n] = r[n] - 2r[n-1] + r[n-2]$

Q&As:

1. $x(t)$, $x[n]$ are periodic signals if

- a) $x(t)=x(t+T)$, $x[n+2N]=x[n]$
- b) $x(t+T)=-x(t+T)$, $x[n+N]=-x[n+N]$
- c) $x(t+T)=x(t)$, $x[n]=x[n+N]$
- d) A+B

Correct: c

2. Communication system which performs the communication process is composed of:

- a) Computers & Wires or Wireless
- b) Hardware & Software
- c) Source & Destination
- d) B+C

Correct: d



3. The goal of communications system is:

- a) Getting the same (sending) data at the destination during acceptable time
- b) Getting the (sending) data at the destination during acceptable time
- c) Getting the same (sending) data at the destination during time
- d) Getting the sending data at the destination

Correct: a

Because communication needs two things

1. Same data → no errors or changes
2. Acceptable time → not too late

4. The transducer converts originated data (voice for example) into signal

- a) Speakers are example of transducer
- b) MIC & digital camera are example of transducer
- c) All input/output devices are example of transducer
- d) B+C

Correct: b

Speakers are output transducers (signal -> sound), while the question asks about converting originated data into a signal, so MIC (voice -> electrical signal) and digital camera (light -> digital signal) are the intended input transducers.

5. Signal is something that carries information this 'something' is 13-Signal is something that carries information this 'something' is physical quantity that can be

- a) Manipulated, Stored or transmitted by analog processes
- b) Manipulated, Stored or transmitted by computers
- c) Manipulated, Stored or transmitted by digital processes
- d) Manipulated, Stored or transmitted by physical processes

Correct d



6. the signal which periodic and determinate is , while the signal which random and non-periodic is:

- a) Power, energy
- b) Energy, power
- c) Even, odd
- d) Odd, even
- e) None

Correct a

7. Periodic signals are always classified as:

- a) Energy Signals
- b) Power Signals
- c) Neither
- d) Random signals

Correct: b

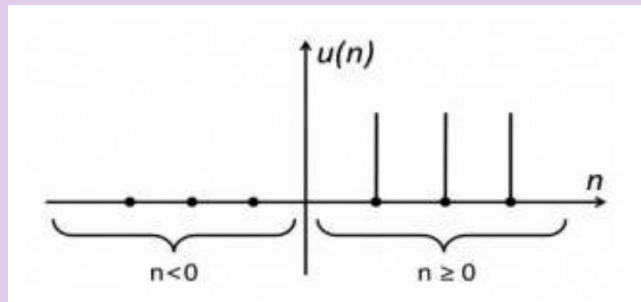
8. For a periodic signal with amplitude A, why is it classified as a Power Signal?

- a) Because its total energy over $(-\infty, \infty)$ is infinite.
- b) Because its amplitude is constant.
- c) Because it has a finite duration.
- d) Because its power is zero

Correct: a

9. The signal in the question below:

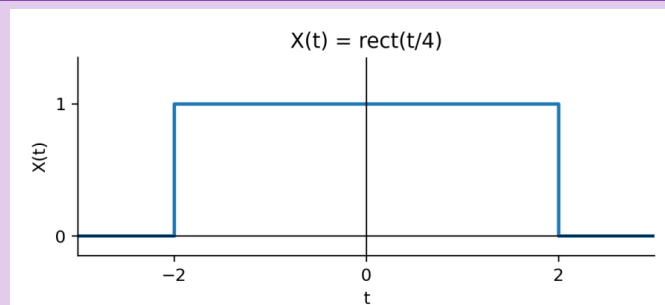
- a) Energy Signal
- b) Power Signal
- c) Even Signal
- d) A and C correct
- e) B and C correct



Correct: a

10. For the below signal where $X(t) = \text{rect}(t/4)$, what is the signal energy E ?

- A. $E = 2$
- B. $E = 4$
- C. $E = 1$
- D. $E = 0$
- E. None



Correct: B

Note: Energy is $E = \int |x(t)|^2 dt = \int \text{from } -2 \text{ to } 2 \text{ of } 1 dt = 4$

11. _____ is a representation of the frequency content of a signal that is composed of sinusoids.

- A. Signal bandwidth
- B. Frequency spectrum
- C. Time domain
- D. Signal spectrum
- E. None



Correct: B

12. What is the frequency F for a sine wave whose period is $1\ \mu\text{s}$?

- A. 50 MHz
- B. 5 MHz
- C. 1 MHz
- D. 10 MHz
- E. None

Correct answer: C

Note: $F = 1/T = 1/(10^{-6}) = 10^6\ \text{Hz} = 1\ \text{MHz}$.

13. For the following sine wave $S(t) = 20 \sin(180\pi t + \phi)$, calculate the signal frequency and the peak-to-peak amplitude.

- A. $F = 1/180\ \text{Hz}$, $A(p-p) = 10\ \text{V}$
- B. $F = 1/180\ \text{Hz}$, $A(p-p) = 20\ \text{V}$
- C. $F = 90\ \text{Hz}$, $A(p-p) = 20\ \text{V}$
- D. $F = 180\ \text{Hz}$, $A(p-p) = 10\ \text{V}$
- E. $F = 90\ \text{Hz}$, $A(p-p) = 40\ \text{V}$

Correct answer: E

Note: Compare $180\pi t$ with $2\pi Ft$: $F = 90\ \text{Hz}$. The amplitude is $20\ \text{V}$, so peak-to-peak amplitude is $2A = 40\ \text{V}$.



14. Given $X(t) = 3.5 \cos(w_1t + 0) + 8.5 \cos(w_2t + 0)$, find the signal power.

- A. $P_X = 42.25 \text{ W}$
- B. $P_X = 65.50 \text{ W}$
- C. $P_X = 36.25 \text{ W}$
- D. $P_X = 42.50 \text{ W}$
- E. None

Correct answer: A

Note: $P = A_1^2/2 + A_2^2/2 = 3.5^2/2 + 8.5^2/2 = 42.25 \text{ W}$.

15. Signal is something that carries information. This “something” is a physical quantity that can be:

- A. Manipulated, stored, or transmitted by analog processes
- B. Manipulated, stored, or transmitted by computers
- C. Manipulated, stored, or transmitted by digital processes
- D. Manipulated, stored, or transmitted by physical processes

Correct answer: D

16. Signal $x(t)$ can be continuous-time. If $x(t)$ can take any value in a continuous interval, then $x(t)$ is:

- A. Discrete
- B. Analog
- C. Random
- D. Digital

Correct answer: B



17. Signal $x[n]$ is defined at discrete times, where n is an integer. Therefore, $x[n]$ is:

- A. Discrete
- B. Analog
- C. Continuous-time
- D. Digital continuous-time

Correct answer: A

18. $x(t)$ and $x[n]$ are even signals if:

- A. $x(-t)=x(t)$, $x[-n]=x[n]$
- B. $x(-t)=-x(t)$, $x[-n]=-x[n]$
- C. $x(-t)=x(-2t)$, $x[-n]=x[-2n]$
- D. A and B

Correct answer: A

19. $x(t)$ and $x[n]$ are odd signals if:

- A. $x(-t)=x(t)$, $x[-n]=x[n]$
- B. $x(-t)=-x(t)$, $x[-n]=-x[n]$
- C. $x(-t)=x(-2t)$, $x[-n]=x[-2n]$
- D. A and B

Correct answer: B



20. $x(t)$ and $x[n]$ are periodic signals if:

- A. $x(t)=x(t+T)$, $x[n+2N]=x[n]$
- B. $x(t+T)=-x(t+T)$, $x[n+N]=-x[n+N]$
- C. $x(t+T)=x(t)$, $x[n]=x[n+N]$
- D. A and B

Correct answer: C

21. $(5t + 3)$ means:

- A. Time delay by 3 and stretching by 5
- B. Time advance by 3 and compression by 5
- C. Time advance by 3 and stretching by 5
- D. Time delay by 3 and compression by 5

Correct answer: B

Note: Exam-style answer. Mathematically, $X(5t+3)=X(5(t+3/5))$, so the exact advance after factoring is $3/5$.

22. The signal which is periodic and deterministic is _____, while the signal which is random and non-periodic is _____.

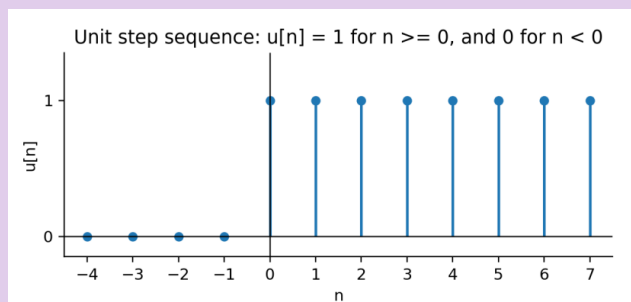
- A. Power, energy
- B. Energy, power
- C. Even, odd
- D. Odd, even
- E. None

Correct answer: A

Note: In this course context, periodic signals are treated as power signals, while non-periodic finite-duration signals are treated as energy signals.

23. The above unit step sequence is classified as:

- A. Energy signal
- B. Power signal
- C. Odd signal
- D. Even signal
- E. None

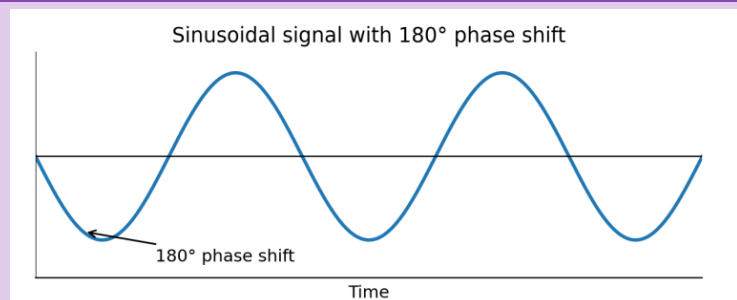


Correct answer: B

Note: Its energy is infinite, but its average power is finite.

24. The following sine wave has a phase shift of 180 degrees. Choose the correct expression.

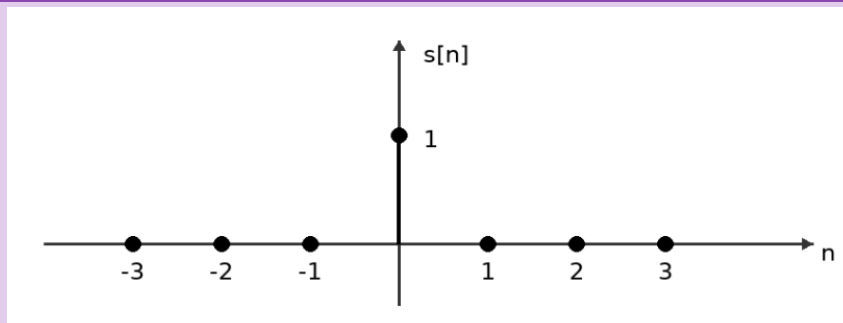
- A. $x(t) = A \sin(2\pi ft + 180^\circ)$
- B. $x(t) = A \sin(2\pi ft + 90^\circ)$
- C. $x(t) = A \sin(2\pi ft - 180^\circ)$
- D. None



Correct answer: A

Note: Mathematically, $+180^\circ$ and -180° produce the same phase inversion, but the expected exam answer here is A.

25. For the below shape, choose the correct signal name.

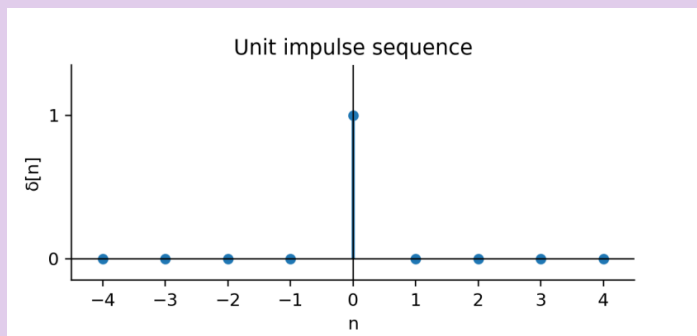


- A. Dirac impulse
- B. Unit impulse operator
- C. Unit impulse sequence for a discrete-time signal
- D. Unit impulse function for a continuous-time signal
- E. None

Correct answer: B

26. The above signal $\delta[n]$ is:

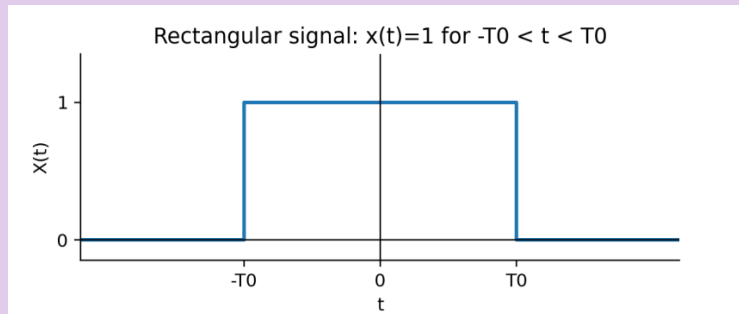
- A. Energy signal
- B. Power signal
- C. Even signal
- D. A and C are correct
- E. B and C are correct



Correct answer: D

Note: The unit impulse sequence has finite energy and is symmetric around $n = 0$.

27. Classify the below rectangular signal.

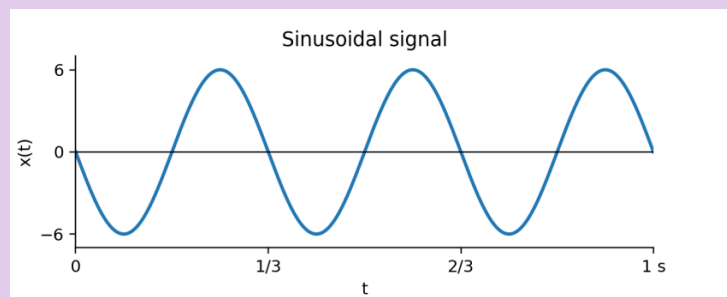


- A. Power signal - even - non-periodic - deterministic
- B. Energy signal - odd - non-periodic - random
- C. Power signal - even - periodic - random
- D. Energy signal - even - non-periodic - deterministic
- E. Energy signal - odd - periodic - deterministic

Correct answer: D

28. For the below sine wave, define (A, F, ϕ).

- A. $A=6$, $F=6$, $\phi=0^\circ$
- B. $A=6$, $F=3$, $\phi=-90^\circ$
- C. $A=6$, $F=3$, $\phi=0^\circ$
- D. $A=6$, $F=3$, $\phi=90^\circ$
- E. None



Correct answer: E

Note: The correct values are $A = 6$, $F = 3$ Hz, and $\phi = 180^\circ$.

The End..

وفقكم الله لا يحب ويرضى! ❤️